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Relationship among Best, Average & worst Case.

- Best Case: minimum execution time when the input is most favorable.
- Avg case: Expected execution time for typical inputs
- worst case: maximum execution time the input is least favorable

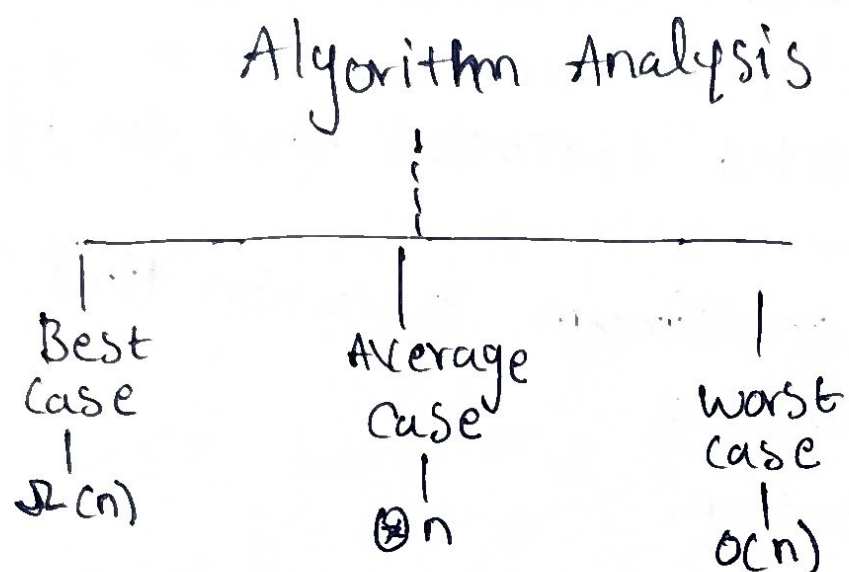
Asymptotic Notation.

- Best Case: Ω : Lower bound
- Avg case: Θ : Tight bound
- worst case: $\text{Big}(O)$: upper bound

Importance:

- Best Case shows the minimum possible performance
- Avg Case represent real-world performance
- worst case guarantees the maximum time an algorithm may take
- Using only one measure does not provide a complete understanding of an algorithm. Considering all three helps compare both practical performance & theoretical efficiency allowing developers to choose the most suitable algorithm.

Concept map



2) Nested Loops & Polynomial Complexity

* Nested loops increases time complexity because one loop executes inside another

- one loop $\rightarrow O(n)$
- two nested loops $\rightarrow O(n^2)$
- Three nested loops $\rightarrow O(n^3)$

* As the no. of nested loops increases, the no. of operations grows rapidly.

Affects Scalability:

- $O(n^2)$ becomes slow for large inputs
- $O(n^3)$ is much slower & unsuitable for very large datasets
- Higher polynomial complexity reduces efficiency & increases execution time

The Concept map helps identify inefficient algorithm designs by showing how additional nested loops increase complexity, making it easier to compare algorithms & choose more suitable solutions

Concept Map:

